

PRODUCT REQUIREMENTS DOCUMENT

ResolveAI

AI-Powered UPI Fraud Evidence & Dispute Resolution Platform

MadeInIndia 2026 Hackathon • Track ZAFD-005

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1. Purpose

This document defines what ResolveAI is, who it is for, what problem it solves, and what the platform must do to succeed at Phase 1 (18 July 2026) and Phase 2 (14 August 2026) of the MadeInIndia 2026 Hackathon. It is the single source of truth the whole team — backend, frontend, and AI — should build against.

2. Problem Statement

OFFICIAL PROBLEM STATEMENT

Victims of digital payment fraud often struggle to gather evidence and navigate complex dispute resolution processes, resulting in delayed outcomes.

When a person is scammed via UPI, three things typically go wrong at once:

- Panic and confusion — the victim does not know what evidence matters or how to organise it.
- Process ignorance — they do not know whom to contact (bank, NPCI, 1930, cybercrime.gov.in) or in what order.
- Poor complaint quality — banks and the cybercrime portal receive vague, unstructured complaints, which slows down investigation and reduces the chance of fund recovery.

ResolveAI's job is to turn scattered evidence (screenshots, SMS, chats) into a structured, chronological, evidence-backed complaint — in minutes, not hours — and to guide the victim through the correct reporting sequence.

3. Goals & Success Metrics

Goal	Metric (for demo / judging)
Reduce time to file a usable complaint	Evidence upload to generated PDF complaint in under 3 minutes
Improve complaint quality	Generated complaint includes all fields a bank/1930 operator requires (UTR, amount, date/time, VPA, description)
Improve evidence completeness awareness	Evidence Completeness Score correctly flags missing critical fields in test cases
Demonstrate real fraud-pattern detection	Correctly identifies fraud type in at least 4 of 5 test scenarios
Judge impact score	Clear story: panic → structured case → bank/police-ready document

4. Target Users / Personas

Persona 1 — Priya, 24, first-time UPI fraud victim

Lost ₹8,000 to a fake customs/parcel scam call. Panicked, does not know a UTR from a transaction ID, has only WhatsApp screenshots and a bank SMS. Needs a simple upload flow leading to a guided complaint, written in plain language.

Persona 2 — Ramesh, 52, small shop owner

Lost ₹45,000 via a fake QR code swap at his shop. Has multiple transaction screenshots across several days. Needs a timeline view to prove the pattern, and a document he can hand to his bank branch in person.

Persona 3 — Bank/NPCI reviewer (indirect user)

Receives the generated complaint. Needs complete, factual, chronologically ordered information with no speculation, so the case can be actioned quickly rather than sent back for clarification.

5. User Journey

Step	Description
1. Login	User authenticates via JWT-based login.
2. Dashboard	User sees a list of their existing fraud cases.
3. Create Fraud Case	User enters basic details — bank, approximate date, approximate amount.
4. Upload Evidence	User uploads screenshots, SMS captures, or chat exports.
5. OCR Extraction	Tesseract OCR converts each image into raw text.
6. AI Extraction	Gemini pulls structured fields — amount, UTR, date, time, VPA, beneficiary.
7. Timeline Construction	Gemini orders all evidence into a chronological sequence.
8. Fraud Indicator Detection	Gemini flags likely fraud pattern(s) and computes an Evidence Completeness Score.
9. Complaint Generation	Gemini drafts a formal, structured complaint document.
10. PDF Report	The complaint is rendered as a downloadable PDF.
11. Case Dashboard	User sees case status, the generated documents, and next-step guidance.

6. Scope

6.1 Phase 1 — MVP (must-have for the 18 July submission)

- User login (JWT authentication)
- Dashboard listing cases
- Create Case
- Upload Evidence (image/screenshot upload)
- OCR extraction (Tesseract)
- AI analysis: field extraction, timeline, fraud indicators, evidence score
- AI-generated complaint text
- PDF download of the complaint
- Basic case dashboard showing status and generated documents

6.2 Phase 2 — Future scope (roadmap to mention to judges)

- Voice input for evidence description

- Multilingual support (regional languages)
- Bank-side dashboard for reviewing submitted cases
- RAG (retrieval-augmented generation) grounded in RBI/NPCI guideline documents, so complaint language cites the correct clause
- Case status tracking, mirroring the App/PSP → Bank → NPCI → Ombudsman escalation
- Advanced analytics on fraud pattern trends

6.3 Out of scope

- Training custom ML/OCR models — the platform uses the Gemini API and Tesseract, not bespoke model training
- Actually submitting complaints to banks/NPCI/cybercrime.gov.in on the user's behalf — the platform prepares documentation; the user still files it
- Payment recovery or fund freezing — this remains with 1930/NPCI/banks; ResolveAI is a preparation and documentation layer, not an enforcement one

7. Functional Requirements

ID	Requirement	Priority
FR1	User can register/log in via JWT-based authentication	Must
FR2	User can create a fraud case with basic metadata	Must
FR3	User can upload one or more evidence images per case	Must
FR4	System runs OCR on each uploaded image	Must
FR5	System extracts structured transaction fields via a Gemini prompt	Must
FR6	System builds a chronological timeline across all evidence in a case	Must
FR7	System detects likely fraud pattern(s) with supporting evidence citations	Must
FR8	System computes an Evidence Completeness Score and lists missing evidence	Must
FR9	System generates a formal complaint document from case data	Must
FR10	User can download the complaint as a PDF	Must
FR11	Dashboard shows all cases with status	Must
FR12	System suggests next steps (e.g. call 1930, file at cybercrime.gov.in, visit bank branch)	Should

8. Non-Functional Requirements

- Accuracy over invention: the AI must never fabricate a transaction field, amount, or fraud claim not present in the evidence. Every extraction prompt must explicitly instruct the model to return null if a field is not found.
- Explainability: every fraud indicator flagged must cite which evidence item supports it — no black-box claims, since this output may inform a real dispute.
- Speed: end-to-end processing (upload to PDF) should complete within 2–3 minutes for a live demo.

- Privacy: uploaded evidence contains sensitive financial data (account numbers, UPI IDs) and should not be logged or exposed beyond the session/case owner.

9. Domain Grounding

The following points, drawn from RBI, NPCI, and cybercrime-reporting research, shape the prompt design, the evidence checklist, and the complaint language.

RBI Zero Liability rule

Where the deficiency lies with the bank or a third party, and the customer reports within 3 working days of notification, liability is zero. Reporting between 4 and 7 working days caps liability at a limited amount. This is why the platform must clearly timestamp when a case is created relative to the fraud date.

NPCI UDIR escalation path

A complaint moves from the App/PSP (3–5 working days) to the bank, and then to NPCI for complex or fraud cases. The case dashboard's status labels should mirror this ladder so the user always knows where their case sits.

1930 helpline and cybercrime.gov.in

The single most time-critical action for a fraud victim is calling 1930 immediately, since funds can only be frozen mid-chain within a short window, often called the golden hour. The generated complaint should therefore always surface a “Call 1930 now” prompt when a case is recent.

Evidence typically required

UTR/transaction reference number, beneficiary VPA/account/merchant name, exact date and time, amount, screenshots of SMS/chat, and a factual description of the incident.

10. AI Architecture Overview

Stage	Model/Tool	Input	Output
OCR	Tesseract	Uploaded image	Raw text
Extraction	Gemini API	OCR text	Structured JSON (amount, UTR, date, time, VPA, bank)
Timeline	Gemini API	Extracted evidence items	Ordered chronological event list (JSON)
Fraud Detection	Gemini API	Timeline	Flagged patterns, confidence, evidence citation, completeness score
Complaint Generation	Gemini API	Full case data	Formal structured complaint text

Full prompt text for each stage is maintained separately in the AI Prompt Architecture and Test Scenarios documents.

11. Risks & Mitigations

Risk	Mitigation
OCR fails on blurry or low-quality screenshots	Test with varied image quality; allow manual field correction as a fallback
AI hallucinates transaction details	Strict “null if not found” instruction; never let the AI infer unseen values
Judges question legal accuracy	Ground every claim in RBI/NPCI sourced material; position as a preparation tool rather than an auto-filing service
Demo evidence looks unconvincing	Build realistic, clearly fictional test scenarios with plausible Indian bank/UPI formatting

12. Open Questions for the Team

- Do we timestamp case creation separately from the fraud occurrence, to support the RBI 3-working-day liability logic?
- Should the Evidence Completeness Score be shown as a number, a traffic-light indicator, or both?
- Do we want a “Call 1930 now” banner for cases created within 24–48 hours of the fraud date?

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